Advanced Optical Fiber Communications

Problem 19: Bit rate and Shannon capacity

An optical transmission channel provides a channel bandwidth of 50 GHz, while the SNR_{dB} is 18 dB. In order to maximize the bit rate, the right modulation scheme shall be found

a) Calculate the bit rate with the following three codes for forward error correction (FEC) and the BER diagram below. What is the maximum bit rate at which modulation scheme?

Name	Overhead (OH)	Pre-FEC BER
Code 1	3%	$2 \cdot 10^{-5}$
Code 2	7%	$1 \cdot 10^{-3}$
Code 3	20%	$3 \cdot 10^{-2}$

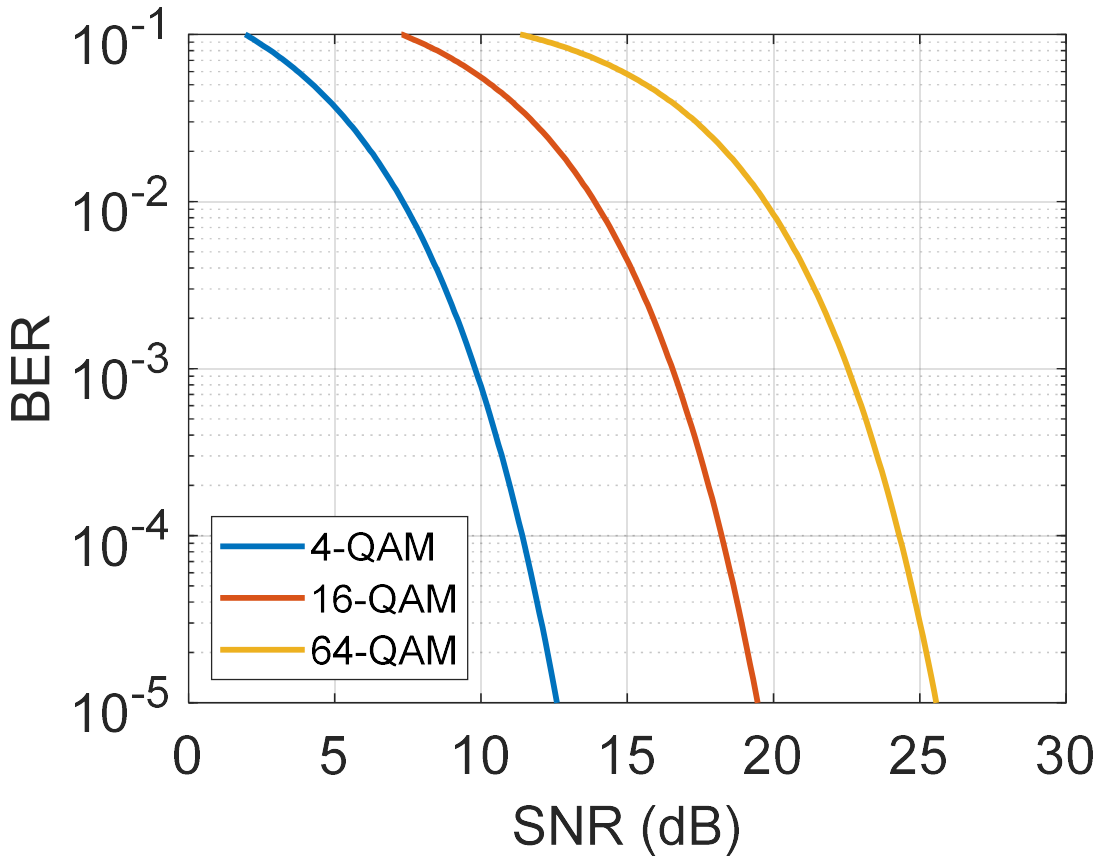
b) Based on the diagram of the Shannon capacity, what is the maximum achievable bit rate of this optical transmission channel?

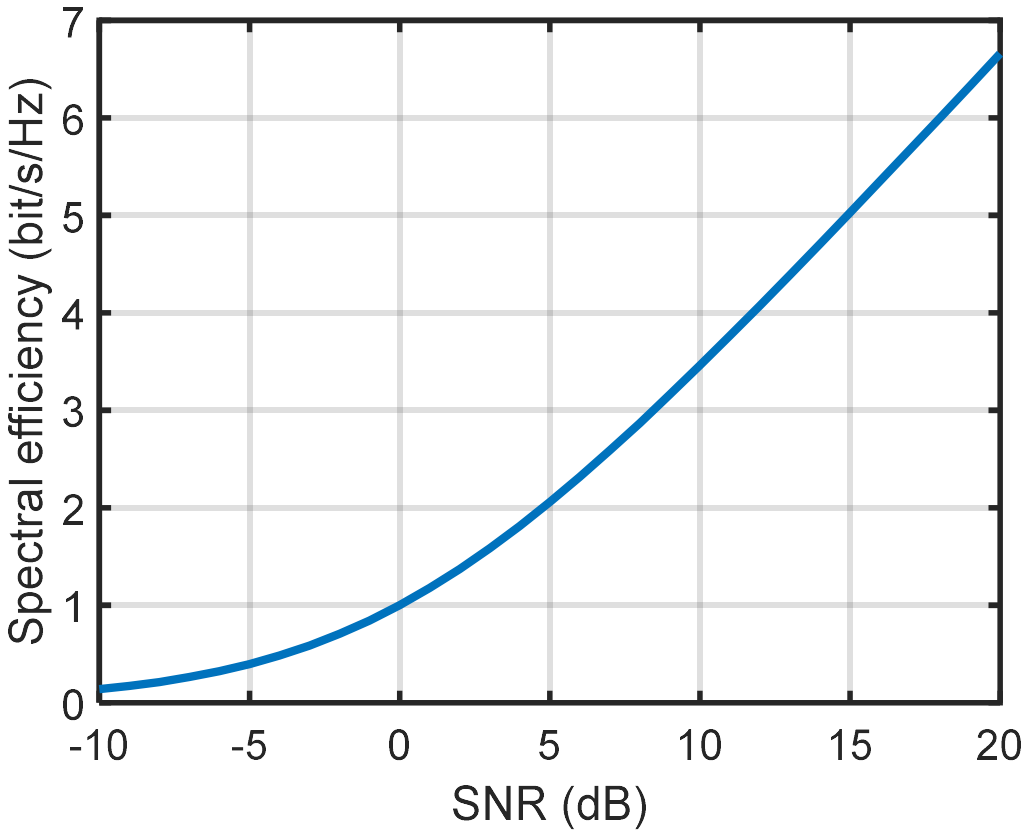
Now, a few-mode fiber with three supported modes is used. The channel matrix is

$$H = \begin{pmatrix} 1 & 0.2 & 0.1 \\ 0.2 & 0.95 & 0.3 \\ 0.1 & 0.3 & 0.9 \end{pmatrix}.$$

c) Calculate the Shannon capacity of the channel. What would be the Shannon capacity without crosstalk?

d) The received signal is $y = (0.3, 1.25, 1.2)^T$. What is the transmitted signal x?


BER
SNR (dB)
4-QAM
16-QAM
64-QAM


Spectral efficiency (bit/s/Hz)
SNR (dB)